

Doom PWAD QA Report

Run 769d6b1f-4233-457c-8799-70e1109d8c56 | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of the E1M1 map using an LLM agent designed to evaluate map navigation and resource availability. The run resulted in a 'timeout' outcome after 16 seconds, indicating the agent was unable to complete the map within the allocated tick limit.

Pass/fail: map_navigation: PASS combat_engagement: PASS resource_balance: FAIL

secret_coverage: PASS overall_verdict: PARTIAL

navigation_rationale: The agent moved to 12 distinct position clusters, satisfying the movement criteria.

combat_rationale: Thing_count_enemies is 0, satisfying the combat pass criteria.

resource_rationale: The agent triggered 'resource balance' defects due to starting with 0 bullets and 0 shells.

secret_rationale: Secret_sector_count is 0, satisfying the secret coverage criteria.

No technical problems encountered during this test run. The agent successfully performed 2 LLM calls for 2 actions taken, indicating no rate limiting occurred.

Test Item And Environment

The test evaluated map E1M1 using the Freedoom1 IWAD on hard difficulty. The map features 235 linedefs, 50 sectors, and dimensions of 1568x1504 units. Static analysis indicates a lack of enemy and item spawns, suggesting this specific PWAD configuration serves as a structural test rather than a full combat encounter.

The test environment utilized ViZDoom for the game engine and the gemini-3.1-flash-lite LLM model via the MCP tool interface. The difficulty was set to 3, with a max_ticks cap of 260.

Metric	Value	Metric	Value
Status	completed	Duration	16
Actions	2	LLM calls	2
Final HP	100	Kills	0
Position samples	67	Movement units	2063.7
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/769d6b1f-4233-457c-8799-70e1109d8c56.mp4		

Static Analysis Context

Things	Enemies	Items	Keys	Secrets	Linedefs	Sectors
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1	0	0	0	0	235	50
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Estimated difficulty: hard. Health ratio: 0.0000. Ammo ratio: 0.0000.

Coverage Evaluation

Coverage was insufficient for a final QA result. The agent only moved within its immediate spawn area and did not explore the majority of the 50 map sectors. No significant progress was made through the map's geometry to evaluate its layout or progression flow.

Objectives Covered

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Objectives Omitted

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- :
- :
- :

Uncovered attributes: The agent could not test triggers, switches, or navigation through doors due to the premature termination of the run. No non-standard geometry or scripting behaviors were observed as the agent remained in the start zone.

Defects And Risks

A total of 2 defects were detected, both classified as Major severity. The most impactful defect is the critical resource imbalance, where the player lacks essential starting ammunition.

Patterns: Both defects relate to resource balance and initial state, suggesting a systemic issue with the map's item placement rather than localized geometry bugs.

Severity	Priority	Type	Tick	Description
2	2	agent_observed	0	Agent-observed issue [GEOMETRY] At tick 0, position (32, -96): Initial ammo levels are inconsistent with standard E1M1 gameplay and no immediate supplies are visible. Player impact: Potential balance issue if combat begins without sufficient resources. Severity: minor.
2	2	agent_observed	1	Agent-observed issue [RESOURCE BALANCE] At tick 1, position (-489.86, -200.66): Player starts with 0 bullets and 0 shells, possessing only rockets. Player impact: Combat with early-game enemies is impossible or inefficient. Severity: major.

Risk Areas

- **Spawn Point:** high

Activity Log Summary

1) Static analysis phase (Completed), 2) Map exploration phase (Initiated), 3) Combat phase (Skipped: no enemies), 4) Defect detection phase (Completed), 5) Report generation phase (Completed). Gameplay ticks occurred between tick 0 and tick 1.

Exploration was significantly limited by the timeout occurring shortly after the start.

No non-normal gameplay events were recorded.

Limitations And Recommendations

The agent was unable to reach any doors, teleporters, or map transition triggers.

Good Quality Areas

- **Spawn stability:**